KRISHNA SHIVHARE

Computer Science Undergraduate • Full-Stack, Agentic AI & Blockchain Development

Nagpur, Maharashtra • krishna12aprill@gmail.com • +91 8263957947 • linkedin.com/in/krishna-shivhare-b03b37282 • github.com/Krishnashivhare

PROFESSIONAL SUMMARY

Undergraduate engineering student at Symbiosis Institute of Technology with a strong technical focus on full-stack web engineering, agentic AI systems, Python-driven applications, and decentralized technologies. Proven competence in building typed tool-calling pipelines, responsive user interfaces, and cryptographic credential flows. Co-author of research on disaster-response resource allocation and GaN RF devices. Experienced in applying probability and statistics to software logic.

TECHNICAL COMPETENCIES

Programming Languages:	Python, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
Web & Frontend Systems:	Next.js, React, Tailwind CSS, DOM Manipulation, Web Storage APIs, Real-Time Validation, Responsive Design, REST APIs, JSON
AI & Agentic Systems:	LLM Tool-Calling, Vercel AI SDK, Zod Schema Validation, Structured Output, Prompt Design
Emerging Technologies:	Blockchain Architecture, Smart Contract Fundamentals, Cryptographic Hashing, Distributed Ledgers, IoT (ESP32)
Core Fundamentals & Tools:	Data Structures & Algorithms, OOP, Probability & Statistics, Simulation & Analysis, Technical Writing, Git/GitHub, VS Code, Vercel

EDUCATION

Symbiosis Institute of Technology (SIT) — Nagpur, Maharashtra **Expected Graduation: 2028**

Bachelor of Technology (B.Tech) in Computer Science / Engineering | Current Standing: Semester 3

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Python), Probability & Statistics, Web Technologies, Database Systems.

KEY PROJECTS

Automated Irrigation Decision Agent **Agentic AI • IoT**

Tech Stack: TypeScript, Next.js 16, React 19, Vercel AI SDK, Zod, Tailwind CSS, Open-Meteo API, ESP32 | github.com/Krishnashivhare/flexi-ca2

- Built an agentic irrigation system replacing a static moisture threshold with a six-phase reasoning loop that returns IRRIGATE, WAIT or MONITOR with litres, pump duration and confidence.
- Designed six Zod-typed domain tools (weather, soil physics, crop knowledge, history, hydrology, decision) with a bounded tool-calling loop, five explicit conflict-reconciliation rules, an auditable water-deficit formula, and a live timeline UI.
- Created a polymorphic sensor interface (simulated or validated ESP32 telemetry) and deterministic fallback engine; verified across 8 functional and failure-path test cases.

Cloud-Based Disaster Response Coordination Platform (Research Paper, Co-author) **Cloud • Decision Support**

Focus: Priority-Based Allocation, Greedy Nearest-Unit Assignment, Random Forest Severity Prediction (proposed), Simulation

- Co-authored a paper on a coordination platform ranking incidents via weighted priority scoring and allocating ambulances, rescue teams, and supplies via greedy nearest-unit assignment with shortage tracking.
- Evaluated allocation across a 40-run simulation of 80 synthetic incidents: cut mean first-response time for Critical incidents from 57.8 to 15.9 minutes versus arrival order, quantifying starvation trade-offs.
- Specified a Random Forest severity classifier (scikit-learn, stratified cross-validation, rule-based fallback) as a proposed extension.

Blockchain-Based Academic Credential Verification Engine **Decentralized Systems**

Tech Stack: Python, Smart Contracts, Cryptographic Hashing, Web Integration

- Architected a tamper-proof prototype to issue, store, and audit student credentials using immutable ledger principles, with direct cryptographic hash lookups eliminating third-party bottlenecks.
- Designed administrative workflow schemas enabling university departments to mint credentials while allowing employers one-click verification.

Dynamic Client-Side Application Suite & Validation Framework **Web Development**

Tech Stack: JavaScript (Vanilla ES6+), HTML5, CSS3, Web Storage API

- Engineered a theme-switching engine using localStorage and sessionStorage for zero-latency UI persistence, plus an event-driven form validation pipeline intercepting invalid submissions before server transmission.
- Implemented dynamic countdown and timing utilities with efficient DOM manipulation patterns and modular state structures.

Investigation of GaN Devices for RF Applications (PBL Group Project) **Research • Semiconductor Analysis**

Focus: GaN HEMTs, Literature Review, Comparative Analysis (Si, SiC, GaAs, LDMOS) | Team of 3, SIT Nagpur

- Analyzed how GaN's wide bandgap, 2DEG and thermal properties drive efficiency, power density and linearity in RF systems (5G/6G, radar, satellite), benchmarked against Si, SiC, GaAs, and LDMOS.
- Reviewed trapping, self-heating and reliability limits and their mitigations (field plates, passivation, GaN-on-SiC, DPD, Doherty tracking); delivered technical report and presentation.

LEADERSHIP & CAMPUS ENGAGEMENTS

Event Lead & Core Organizing Member — Flagship & Inter-College Summits **SIT Nagpur | 2024 – Present**

- **Flagship Operations (Enthusia 3.0 & 4.0):** Served in core planning and on-ground execution for annual flagship summits over the past year; managed live-stage flows, event scheduling, and resource coordination.
- **Inter-College Outreach & Logistics:** Coordinated registration, venue orientation, and communications for visiting external collegiate contingents and delegates, managing smooth crowd movement and operations.
- **Internal Hackathons & Sprints:** Directed venue logistics and team onboarding for SIT Nagpur's internal hackathons, leading volunteer teams of 15+ members to maintain rigorous event schedules.

Competitive Hackathon & Technical Event Participant **Inter-College & Institutional | 2024 – Present**

- **Rapid Prototyping:** Represented SIT Nagpur across internal and external hackathons, collaborating in cross-functional teams to build sprint-based full-stack software and AI prototypes under 24–36 hour deadlines.
- **Collaborative Development:** Managed shared Git repositories, modular branch merges, and technical demo presentations during competitive builds.